Finding Optimal Q-Values

45 minutes

Preparation and materials

1. Slides 1–21
2. Board, paper, or iPad
3. charger_grid.py

Before class · Gymnasium reading (10 minutes). Read Farama's Gymnasium Basic Usage guide from "Why Gymnasium?" through "Explaining the Code Step by Step." Focus on reset() and step(action). Skim the CartPole example without running it.

Instructor setup: install Gymnasium, run the two-episode demo, and have the slide 19 results ready to project.

Learning outcomes

Use tabular Q-learning to estimate the optimal Q-values for the charger grid: the best discounted return for each starting cell and first move, assuming the robot makes the best choices afterward.

1. Identify which Q-table entry belongs to a given cell and move.
2. Update estimates by hand using the Q-learning formula.

   New Q = Old Q + α × (Target − Old Q)
   Target = Reward + γ × best Q-value in the next state
   If the episode ends, Target = Reward.

3. Run the same update in Python. Use the learned table to pick a move and explain how learning from rewards is reinforcement learning.

Lesson stage	Time	Activities
Bridge In	0–2 min	Slide 1. Show the robot and charger. Ask which first move gives the robot the best chance of charging sooner, and how we could put a number on that move.
Learning Outcomes	2–3 min	Introduce tabular Q-learning as a way to estimate the best discounted return for each move from experience.
Pre Assessment	3–6 min	Slides 2–4. Ask pairs to trace the two shortest routes from A to D and predict whether their first moves should have equal optimal Q-values.
Participatory Learning	6–18 min	Slides 5–9. Use +10 at D and zero elsewhere. Compare the direct route from B with a detour to show why discounting favors the earlier reward. Build the zero-filled Q-table and predict the best return from A for each of the two shortest routes.

Lesson stage	Time	Activities
Participatory Learning	18–28 min	Slides 10–15. Update A/Right and B/Down by hand for two episodes. Compare the estimates 2.25 and 7.5 with the optimal values 9 and 10. Ask why two visits have not yet produced the optimal values.
Participatory Learning	28–32 min	Slides 16–17. Trace the route through C. Ask whether a zero for A/Down means the move is worse or that the agent has not explored it. Both shortest routes from A give an optimal value of 9.
Participatory Learning	32–37 min	Slides 18–19. Write the update line and run charger_grid.py in Gymnasium. Compare row C with and without exploration. Right from C has an optimal value of 10, but the 20-episode runs never tried it.
Post Assessment	37–41 min	Slide 20. Ask learners to calculate the target and updated Q-value for B/Down when old Q = 7.5 and α = 0.5. Check that the target is 10 and the update is 8.75. Ask which of these is the optimal value.
Summary	41–42 min	Use the learned table to pick a move from A. Then compare with the optimal values: Right and Down both give 9. Name the method: tabular Q-learning is reinforcement learning.
Summary	42–45 min	Slide 21. Preview how a DQN estimates the same action values for the current state when a larger grid makes a table less practical.